Purushotham Sirasapalli

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Education

University at Buffalo — Buffalo, NY
Bachelor of Science in Computer Science, Minor in Mathematics | GPA: 3.50/4.00 — *Aug 2022 – May 2026*

Experience

Software Engineer, SCoRe Lab — Jan 2025 – May 2026
University at Buffalo — *Buffalo, NY*

- Built Coriolis-Lite, a memory-efficient exact-match search engine in C++ that uses compressed text indexes (FM-indexes) to classify biological sequences on mobile devices under strict memory limits.
- Rebuilt the caching layer of SKiM in Rust using Tokio and rayon, streaming from SSD with key-level deduplication, for a 6× speedup and 80% less RAM (4GB to 800MB) on a 63M-key benchmark.
- Replaced SKiM's hash-based index with minimal perfect hashing, removing the hash table's per-key overhead on a 63M-key reference database.

Undergraduate Research Assistant, ADAMS Lab — Dec 2024 – May 2026
University at Buffalo — *Buffalo, NY*

- Built a decentralized swarm of TurtleBot3 robots that locate radio-frequency sources with no central coordinator. Deployed on **4 physical robots** (ROS2 Humble on Raspberry Pi 4) and validated with up to **50 simulated agents**.
- Implemented a Bayesian-Optimization + Gaussian-Process inference pipeline for the swarm that, across **99 physical** and 194 simulated experiments, located the source in **83% of physical runs** and within mission time on all but 11 of the 194 held-out scenarios.
- Trained a reinforcement-learning policy for vision-based aerial target tracking, raising stable tracking speed from 2 m/s to **10 m/s** versus the prior PID-based controller.
- Cut on-robot GP inference time on the Raspberry Pi from **3–4 minutes** to a few seconds by running a small neural network that compresses sensor observations to **100 points** while preserving accuracy, enabling real-time decentralized tracking.
- Designed a bi-level optimization framework that jointly tunes the swarm's search behavior and individual UAV hardware, with the co-designed system finishing missions **44% faster** than a sequential design in 194 simulated scenarios based on avalanche rescue.

Student Manager, 3 Pillar Catering — Aug 2023 – May 2026
University at Buffalo — *Buffalo, NY*

- Promoted from student staff to Student Manager; led a team of 10+ student staff in a high-volume catering operation, managing shift logistics and onboarding for events of up to **1,000 guests**.

Software Engineering Intern — Feb 2022 – May 2022
Foundry Medical Technologies — *Visakhapatnam, India*

- Implemented and validated embedded control logic for a medical ventilator across adult and pediatric modes, conducting mode-level hardware testing to verify compliance with safety and performance standards.

Projects

Decentralized Signal Source Localization for Swarm Robots | *ROS2, Python, PyTorch*

- Integrated LoRa RF sensors (20 Hz, 868 MHz) across a 4-robot TurtleBot3 swarm with median absolute deviation and local neighborhood filtering, supporting decentralized source localization across **99 physical experiments** with 83% mission success.
- Built the ROS2 Humble communication and sensor integration stack on TurtleBot3 + Raspberry Pi 4 (Ubuntu 22.04 Server), integrating Vicon 12-camera motion capture at 20 Hz and enabling decentralized waypoint planning and collision avoidance across the swarm.

Technical Skills

Languages: C/C++ · Python · Rust · Java · JavaScript · SQL
Frameworks & Tools: PyTorch · CUDA · Stable-Baselines3 · OpenAI Gym · scikit-learn · NumPy · ROS2 · Gazebo · Hadoop · Spark · Docker · React · Git · Linux
Cloud: AWS (EC2, S3, Lambda, IAM, CloudFormation, CloudWatch)